Project 2- Histograms

Part 1—Histogram Equalization

The histogram equalization brightens dark images and darkens bright images quite effectively, with the second user-submitted photo being the best example of the brightness reduction. The from-scratch equalization (second row) is clearly identical to the cv2 equalization (third row).

Part 2—Histogram Matching

The histogram matching method clearly depends much more on the chosen image. The contrast is increased in the given images thanks to the high-contrast leaf, and image 3 has a similar effect. Source image 4 is fascinating because the bleed-through of the pavement outside is more prominent than it is in the equalization, but I believe this is because the darkening is more drastic.

Here, the from-scratch method can be seen to be identical to the SciKit histogram matching function.

Part 3—Thoughts on Color Extension

This would likely produce some strange results in just RGB channels because they are often not directly correlated with luminance. HSV and LAB color systems would have better results because they relate directly to luminance.